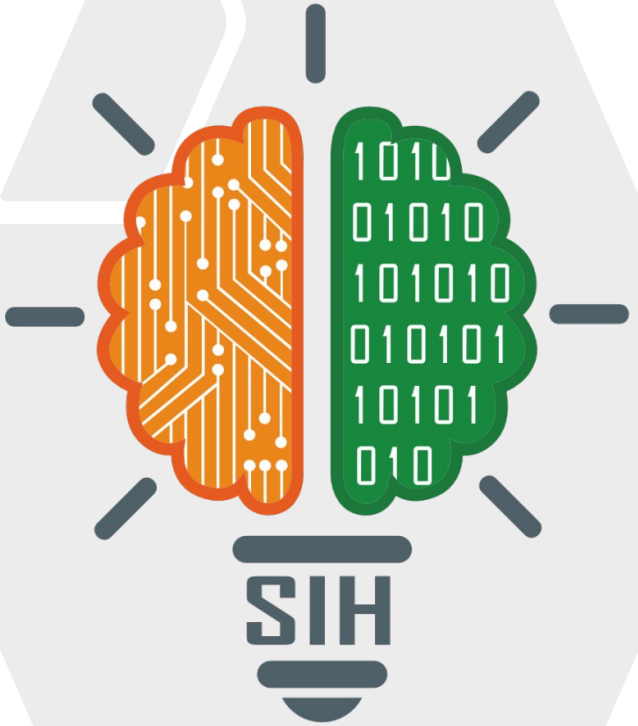
# SMART INDIA HACKATHON 2026

# SCOURGE

*Ask the bank, not the paper.*




- **Problem Statement ID –** SIH26265
- **Problem Statement Title –** Streamline of Security Deposit / FDR Portal
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** 199322
- **Team Name (Registered on portal) –** Arnavax#

SCOURGE

# SCOURGE — ASK THE BANK, NOT THE PAPER

SMART INDIA HACKATHON 2026

## Proposed Solution (Describe your Idea/Solution/Prototype)

**The bank's own website proves the FDR: no bank API, no agreement, no paper. AICTE sees only that record.**

### Detailed explanation of the proposed solution

AICTE sends a **single-use challenge** for one FDR. The institution opens its own net banking; the connection key is **split with AICTE's verifier** (MPC-TLS), so nobody can fake the page. Login, OTP and PAN stay with the institution.

### How it addresses the problem

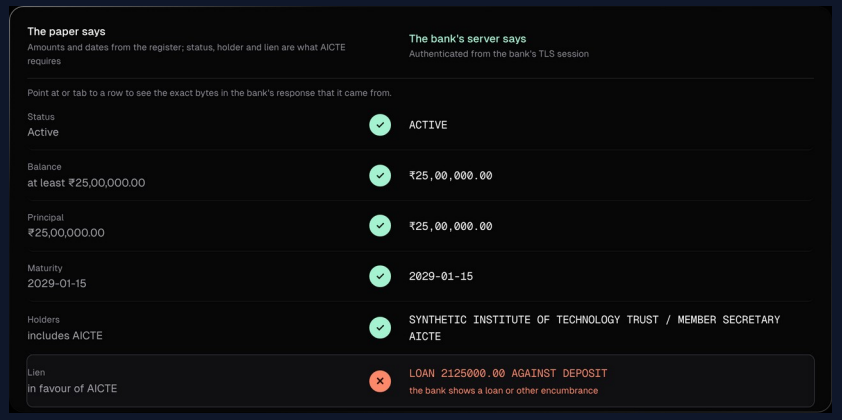
Paper is checked once and trusted for 10 years. Scourge **re-checks the bank any time and before every release**: a loan, early encashment, missing pledge (lien) or fake FDR shows up as a **Discrepancy**.

### Innovation and uniqueness of the solution

**Works with banks' existing net banking**: no bank API or agreement needed, and API-ready when banks offer one. If a page cannot show a field: bank-signed e-advice, then branch confirmation. Fakes and replays are rejected.

LIVE PROTOTYPE · SYNTHETIC BANK

**The bank's server contradicts the paper.**




× HIDDEN LOAN CAUGHT

Every value traces to the bank's own bytes

## The complete FDR lifecycle in one portal (all five stages run in the prototype)

**1 FILE**
Institution files online; renewals linked

→

**2 PROVE**
Bank-session proof on AICTE's challenge

→

**3 MONITOR**
T-90/60/30/7 alerts; adequacy; MT940/camt.053 reconcile

→

**4 DECIDE**
Verdict + evidence; hash-chained audit

→

**5 PLEDGE RELEASE**
Dept reviewers, 2 signers, holds; signed letter → bank ack

**Why now:** **₹201.42 crore of FDRs "do not appear in IDFC's records"** (CBI chargesheet, Oct 2026); AICTE joint FDRs found encashed or loaned against without consent.

SCOURGE

# TECHNICAL APPROACH

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## Technologies to be used

**Proof layer**
TLSNotary MPC-TLS · Rust
Selective disclosure · rustls

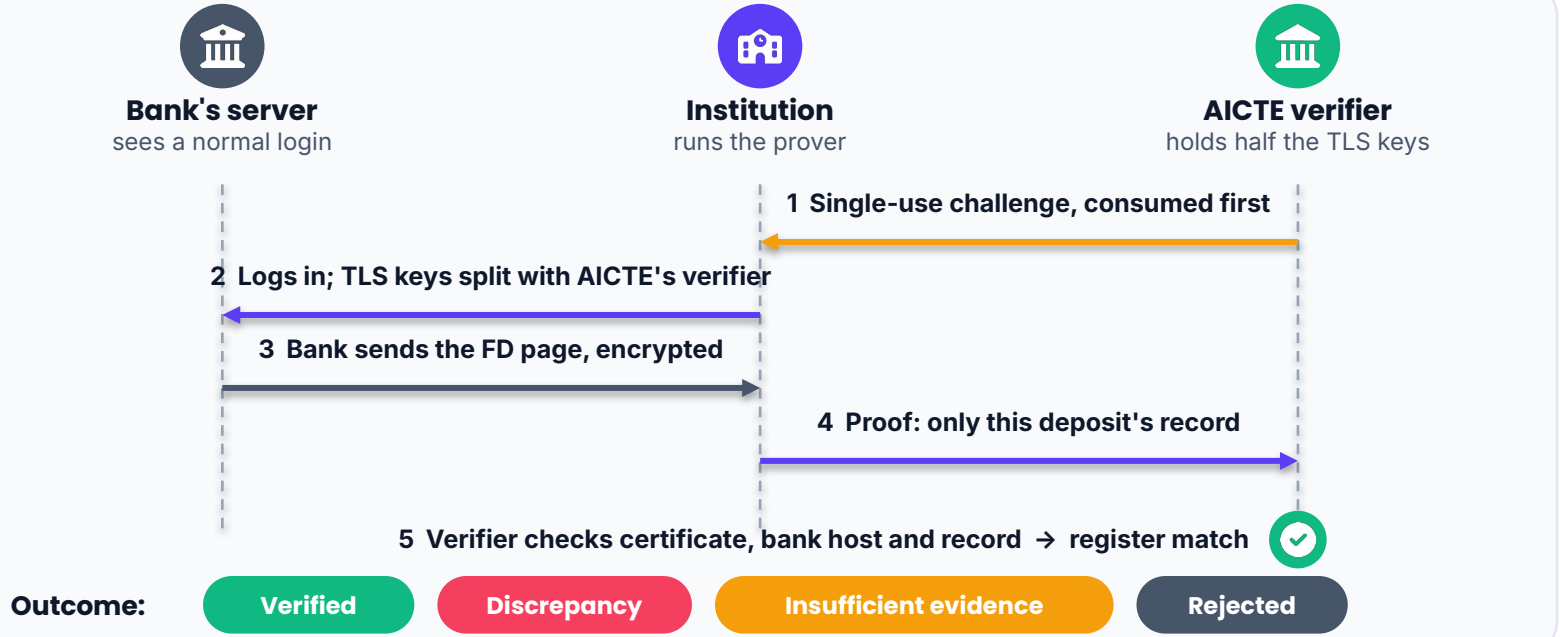
**Backend**
Python · FastAPI · SQLite
AES-GCM vault · hash-chained audit

**Frontend**
Server-rendered HTML · JS
Live 3-party proof view

**Ops & QA**
Docker · nginx · GitHub CI
298 tests · 2FA · audits

## Methodology and process for implementation (Flow Charts/Images/ working prototype)

**In plain words:** *an AICTE "notary" watches the bank page load, sees only the FD line, and nobody can change it.*




**WATCH THE 3-MINUTE DEMO**




**Click to play here, or scan / click the QR for YouTube**
Synthetic bank + live real-bank session

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# FEASIBILITY AND VIABILITY

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## Analysis of the feasibility of the idea

**0.58 s**
**per bank proof**
*median of 60 runs · synthetic bank*

**Real bank**
**live split-key session**
*logged-in deposits API · 200 OK*

**298 / 298**
**automated tests pass**
*incl. rollback, forgery, backup-restore*

**₹0**
**software licences**
*60 MB/proof (measured) · FEASIBILITY.md*

**LIVE ON A REAL BANK**
First bank tested: SBI · 1 Nov 2026

{"ok":true,"server":"api.onlineyono.sbi.bank.in","status_line":"HTTP/1.1 200 OK","secs":0.4}

*Split-key session with a bank's logged-in deposits API (first bank tested: SBI); cert checked against public roots. Log: tools/realbank/RESULTS.md*

## Potential challenges and risks

## Strategies for overcoming these challenges

| | |
|---|---|
| Bank page can't be proven, or hides the lien | 3-tier fallback: bank-signed e-advice, then branch; never a guess |
| Fake bank site or a replayed old proof | Rejected: bank certificate + single-use, 15-min challenge |
| Bank terms, consent and data protection | Own login, per-deposit consent, record-only disclosure; legal review |
| Bank encrypts data inside HTTPS, or ties it to the browser | Browser-extension prover; verifier gets the per-session app key |

**PILOT PLAN** ✓ Prototype, 298 tests → ✓ Live real-bank session → FD pages at 3 large banks → 1 AICTE regional office → National rollout

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# IMPACT AND BENEFITS

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## Potential impact on the target audience

### AICTE RB officers
A dated proof from the bank in seconds, not a letter in weeks

### AICTE finance
The whole FDR book is re-checkable, before every release

### Institutions
One short proof from their own net banking; Hindi or English

### Students
29.6 lakh enrolled in 2025-26 (Lok Sabha reply); deposits checked all 10 years

## Benefits of the solution (social, economic, environmental, etc.)

| TODAY | WITH SCOURGE |
|---|---|
| Branch letter or uploaded screenshot | **The bank's own server, proven** |
| Checked once, trusted for 10 years | **Re-checked any time, before every release** |
| Weeks per confirmation | **~1 s per proof (synthetic, local); ~10 s at 50 Mbit/s** |
| Hidden loans found at release, or never | **Pledge checked at the bank; two-officer release** |

**TESTED ON 180 FRESH DEPOSITS**
Random amounts, dates, formats · real proofs · synthetic bank

| | |
|---|---|
| **119 / 119** | bad deposits the upload portal approved |
| **0 / 119** | bad deposits Scourge verified |
| **0 / 61** | honest deposits Scourge accused |

**Social**
students' deposits stay intact

**Economic**
stops loss before release

**Environmental**
no paper letters or couriers

**Governance**
accounts, 2FA, vault, chained audit

SCOURGE

# RESEARCH AND REFERENCES

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## Details / Links of the reference and research work

### The problem is real

CBI: ₹201.42 cr of FDRs not in IDFC's records (ThePrint, 7 Oct 2026)
theprint.in › cbi-says-haryana-pollution-board...

CBI on AICTE: joint FDRs encashed or loaned without consent
deccanherald.com › greater-transparency-aicte...

### Regulation

AICTE guidelines 2024-27: cl. 6.2(g) 10-year release; cl. 11.8
indiankanoon.org/doc/173919027

Account Aggregator users must be regulated FIUs (DFS)
financialservices.gov.in › account-aggregator...

### Technology

TLSNotary: MPC-TLS proofs of web data, MIT / Apache-2.0
tlsnotary.org

NeSL e-BG for security deposits: new instruments only (WDRA)
wdra.gov.in › ebg_requirement-27-08-2025

**PROJECT RESOURCES**

**Demo video:** youtu.be/8HoplhFOGCg
**Live demo:** scourge-sih26265.onrender.com · 298 tests · synthetic bank
**Code + evidence:** github.com/Lakki007/Arnavax-Scourge-SIH26265

## Scourge vs existing approaches

| | Upload & approve | Bank letter | Account Aggregator | NeSL e-BG | Scourge |
|---|---|---|---|---|---|
| Uses the bank's own data | ✗ | ✓ | ✓ | ✓ | ✓ |
| No bank API or agreement | ✓ | ✓ | ✗ | ✗ | ✓ |
| Institution cannot forge it | ✗ | ~ | ✓ | ✓ | ✓ |
| Covers existing FDR book | ✓ | ✓ | ✓ | ✗ | ✓ |
| Re-check in seconds | ✗ | ✗ | ✓ | ✓ | ✓ |
| Shown on a real bank today | ✓ | ✓ | ✓ | ✓ | ✓ Live bank session |

*~ a bank letter can still be forged. Live split-key session with a real bank's logged-in deposits API: 200 OK (first bank tested: SBI); deposit fields not yet read.*